

Aanal Kamleshkumar Patel

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ON | Canada | [Publication \(Imposter syndrome comparison map\)](#) | Open work permit valid to February 2029

SUMMARY

Embedded AI & Applied AI Engineer with **3+ years** deploying real-time perception and control systems on resource-constrained hardware — **ESP32, Jetson Nano, Raspberry Pi**, and **ARM Cortex** — including **OTA firmware** updates, sensor fusion, and **TinyML** model optimization for on-device inference. Extends that embedded foundation into full-stack **applied AI**: production **LLM/RAG pipelines**, cloud backends, and clinical automation — including Mount Sinai's **chatDATA**, reducing **176 hours of annual manual work to under 64 minutes** across **11 acute-care sites (AMMI 2026)**. Full stack, hardware to cloud: sensor fusion → model building → optimization → embedded deployment → backend & cloud services.

EXPERIENCE

Applied AI Engineer | Lambton Research and Innovation | Contract | Toronto, ON | *June 2025 - Jan 2026*

- Deployed **YOLOv8**-based real-time object detection on **Jetson Nano** and **Raspberry Pi** for a smart inventory system — integrated **LLM**-based semantic labeling pipeline for contextual inventory classification at the edge
- Applied **Post-Training Quantization (PTQ)** to reduce model footprint by **40%**, enabling low-latency inference on **ARM**-based embedded hardware under strict compute constraints
- Designed and deployed a bilingual **RAG**-based AI chatbot using **Azure Data Factory**, **OpenAI GPT**, and **Azure CLU**, achieving **95% system accuracy** through strict similarity gating and contextual retrieval
- Built an **Azure**-hosted PDF ingestion and embedding pipeline with automated chunking, vector embeddings, and semantic retrieval for multilingual document QA
- Developed and containerized **Flask** inference APIs with **Docker**, deployed to **Azure Container Instances (ACI)** with secure RESTful endpoints

AI Developer | RewardsX Loyalty Platform International Inc. | Co-op | Toronto, ON | *January 2025 - April 2025*

- Built a multimodal **computer vision** and **OCR** pipeline, improving extraction quality through data cleaning, confidence scoring, and targeted tuning
- Integrated **Firestore (NoSQL)** for OCR pipeline data storage, supporting structured extraction results across the document processing workflow
- **Containerized** services with **Docker**, supporting CI/CD-style releases and reproducible dev environments
- Contributed **HTML/CSS/TypeScript** front-end support beyond core AI scope, assisting the team's website as a cross-functional contributor

Associate AI Developer | Credwise Financial Inc. | Internship | Toronto, ON | *May 2024 - December 2024*

- Built a time-series sales prediction model using **BigQuery (SQL)** for feature engineering and historical trend analysis
- Contributed to an **LLM/RAG** assistant, improving retrieval quality and adding structured evaluation checks to reduce quality drift
- Implemented backend tooling **APIs (JavaScript/Python)** to integrate model outputs into product flows
- Optimized **GitHub Actions** workflows, reducing deployment friction and improving release velocity

Embedded AI Engineer | Prayosha Food Services Pvt Ltd | Permanent | Ahmedabad, India | *March 2022 - July 2023*

- Architected real-time obstacle avoidance pipeline using **Intel RealSense depth cameras** and **MobileNetV3 CNNs**, cutting navigation latency by **20%** in dynamic indoor environments
- Deployed **TinyML** facial recognition on **ESP32-CAM**, achieving on-device identity verification under strict memory/compute constraints
- Built hybrid indoor navigation integrating depth sensing, IR cameras, and ceiling-mounted **QR tracking** for robust localization
- Implemented secure **OTA firmware updates** across distributed microcontrollers, improving uptime and enabling safe field iteration
- Engineered cloud-integrated telemetry monitoring with automated fault alerts for remote diagnostics

AI ML Intern | GIRO FinTech | Internship | Ahmedabad, India | *July 2021 - October 2021*

- Built a real-time gesture recognition pipeline using **CNN**-based classification, tuning preprocessing and thresholds to reduce false positives
- Ran iterative evaluation loops, improving model stability under varied lighting/motion conditions
- Implemented anti-cheat detection logic using ML signals, improving robustness of predictions

EDUCATION

Lambton College, Toronto, Ontario | Post-Graduate Certificate, Artificial Intelligence and Machine Learning
October 2023 - April 2025

Gujarat Technological University, Gujarat, India | Bachelor of Engineering, Computer Engineering
August 2017 - July 2021

TECHNICAL SKILLS

- **Embedded & Robotics:** ESP32-CAM, Jetson Nano, Raspberry Pi, Intel RealSense, ARM Cortex, Arduino, BLE, OTA Firmware, PID Control, I2C, SPI, UART
- **Edge AI & Model Optimization:** TinyML, TensorFlow Lite, PTQ, INT8 Quantization, CUDA, TensorRT, Knowledge Distillation, Pruning, MobileNetV3, YOLOv8
- **Computer Vision & ML:** Object Detection, Facial Recognition, OCR, CNNs, OpenCV, MediaPipe
- **Languages:** Python, C, C++, Embedded C, JavaScript, TypeScript, HTML/CSS
- **LLM / Applied AI:** RAG Architecture, LLM Tool-Calling & Orchestration, Claude API, OpenAI GPT, Azure CLU, Prompt Engineering, Vector Embeddings, Semantic Retrieval
- **MLOps, Cloud & Data:** Docker, GitHub Actions, Azure (ACI, Data Factory), GCP, BigQuery, Firebase (NoSQL), Flask, RESTful APIs, Git
- **Currently Learning:** ROS2, FreeRTOS

PROJECTS

- **Spacecraft CAN Bus Health Monitor** — Simulated 4-node spacecraft bus (Power/Attitude/Thermal/Propulsion) with FSM-based fault detection, processing **1,868 telemetry frames** and identifying **15 injected faults** in real time.
- **Classical CV Wire Defect Inspection** — Built a class-based defect detection pipeline (**OpenCV**) for industrial wire inspection, from raw differencing prototype to production-structured inspector class.
- **AI Model Performance Benchmarking** — Real-time **edge AI** benchmarking system (**OpenCV, MediaPipe**) with a multithreaded **PySide6** dashboard measuring latency/FPS trade-offs across resolution and power states.
- **Audio Toxicity Detection** — Deep learning model (**TensorFlow/Keras**) for real-time toxic speech detection, validated on live audio inference.
- **ATS Resume Screening System** — **NLP** pipeline using **Named Entity Recognition** and keyword matching to automate resume-to-JD skill alignment.

PUBLICATIONS

chatDATA | Mount Sinai Hospitals | Toronto, ON | *November 2024 - Present (Ongoing)*

- Built an LLM-powered antibiogram automation platform using **Claude API** with tool-calling and prompt orchestration — reduced annual antibiogram generation from **176 hours** of manual expert work to **under 64 minutes** across **11 acute-care sites**; validated and presented at **AMMI 2026**. Served as primary **PHIPA compliance** lead, enforcing PHI data handling and Canadian data residency requirements.

VOLUNTEER WORK

- **Tech Lead** | Yiddishland California | Jan 2025 – Aug 2025 — Improved workflow efficiency by **20%** through delegation and prioritization; maintained **100% adherence** to project timelines.
- **Research Team Member** | Robotics Club, VIER | March 2017 – July 2021 — Embedded programming, ML in embedded systems, **I2C/SPI/UART** protocol implementation.